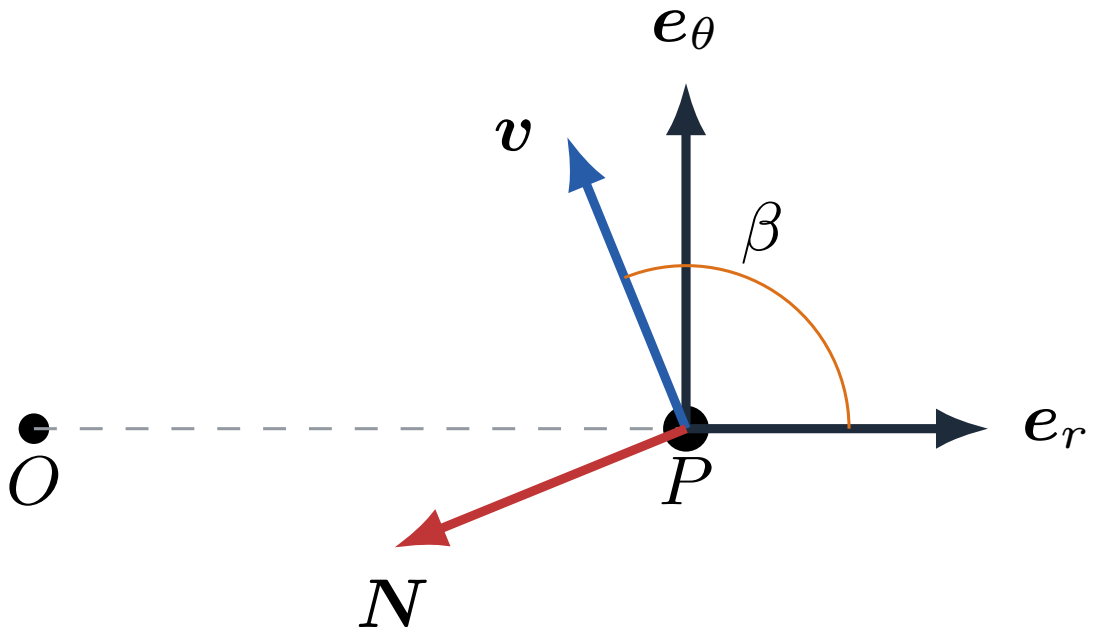

$$\boldsymbol{v} \parallel -a\boldsymbol{e}_r + \boldsymbol{e}_\theta$$

$$\boldsymbol{N} \parallel -\boldsymbol{e}_r - a\boldsymbol{e}_\theta$$

$$\dot{L}_z = -amv_0^2/(1+a^2)$$